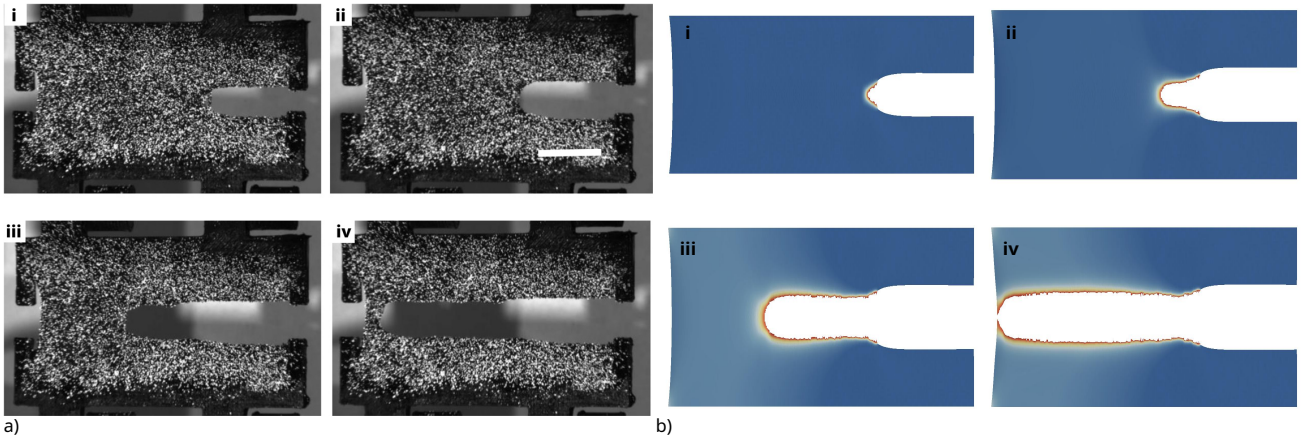
ORTA DOĞU TEKNİK ÜNİVERSİTESİ
MIDDLE EAST TECHNICAL UNIVERSITY

Computational Solid Mechanics Lab (CSML)
Department of Civil Engineering
Middle East Technical University (METU)

Master Thesis (3)

Modeling of Blood Clot Fracture in Human Arteries and Veins

Overview



Mode-I fracture experiment on a blood clot specimen a) speckled DIC images, b) phase-field evolution showing cracking at four time instants.

One in four people die from blood clots that form and then embolize (break off) on the luminal side of arteries and veins. Once broken off, clots travel downstream, where they occlude vital blood vessels such as those of the brain, heart, or lungs, leading to strokes, heart attacks, or pulmonary embolisms, respectively. Despite clots' obvious importance, much remains to be understood about clotting and clot embolization. This fundamental fracture mechanics understanding applies to traditional engineering materials such as steel and aluminum and also to soft materials such as hydrogels, soft tissues, and blood clots. However, exactly which forces impart enough strain energy to exceed clots' fracture toughness is unclear. On top of that, the relationship between external forces and clots' mechanical behavior is still untangled. In fact,

- fibrins in blood clot tend to exhibit strain induced anisotropy as they reorient themselves
- blood clots are essentially a porous media, necessitating the assessment of the effect of interstitial blood pressure

The aim of this project is to **model the strain induced anisotropy and poro-elastic fracture of blood clots** in a holistic and computationally efficient manner by means of a variational phase-field framework that incorporates fibrin realignment and porosity of clots. The expected outcomes include

- a predictive computational tool validated against experimental data
- a better understanding of causes and mechanisms of embolization

Tasks

- extending the already existing isotropic constitutive model (e.g., Ogden model) to include strain induced anisotropy
- implementing the strain-induced material model as a subroutine
- running verification and validation tests for the new model
- preparation of research findings for journal publications

Your qualifications

- B.Sc. degree holder in Civil Engineering, Mechanical Engineering, or related discipline
- strong motivation to learn continuum and computational solid mechanics
- programming experience (e.g. Fortran, Python) and familiarity with FEA tools are two pluses
- good command of written and spoken English

We offer

- direct research mentorship and guidance throughout the thesis work
- access to computational infrastructure and research facilities

Contact

Interested candidates should send their CV & Letter of Interest to:

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